Reflectivity and Transmissivity

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Lecture Objective

- By the end of this lecture, you will be prepared to:
  - Describe the relationship between reflectivity and transmittivity and the Fresnel reflection and transmission coefficients.
  - Compute reflectivity and transmittivity at the planar boundary between two half-spaces comprising different media.

Reflectivity and Transmissivity

- Reflection and transmission coefficients are ratios of reflected and transmitted electric field amplitudes to the amplitude of the incident electric field.
- Reflectivity and transmissivity relate the power ratios of the reflected and transmitted waves to the incident wave.
- In optics, reflectivity is called reflectance, and transmissivity is called transmittance.

Reflection and Transmission Coefficients (1 of 2)

- The reflection coefficient is the ratio of the reflected electric field to the incident electric field:

$$\Gamma_{\perp} = \frac{E_{\perp 0}^{r}}{E_{\perp 0}^{i}} = \frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$$

$$\Gamma_{\parallel} = \frac{E_{\parallel 0}^{r}}{E_{\parallel 0}^{i}} = \frac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$$

Reflection and Transmission Coefficients (2 of 2)

- The transmission coefficient is the ratio of the transmitted electric field to the incident electric field:

$$\tau_{\perp} = \frac{E_{\perp 0}^{t}}{E_{\perp 0}^{i}} = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$$

$$\tau_{\parallel} = \frac{E_{\parallel 0}^{t}}{E_{\parallel 0}^{i}} = \frac{2\eta_2 \cos\theta_t}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$$

Poynting Vectors for Incident, Reflected, and Transmitted Waves

- Poynting (average power density) vectors for incident, reflected, and transmitted waves (either perpendicular or parallel polarization):

$$\vec{S}^i = \hat{k}_i \frac{\left|E_0^i\right|^2}{2\eta_1}$$

$$\vec{S}^t = \begin{cases} \hat{k}_t \dfrac{\left|E_0^t\right|^2}{2\eta_2}, & \theta_i < \theta_c \\ 0, & \theta_i \geq \theta_c \end{cases}$$

$$\vec{S}^r = \hat{k}_r \frac{|E_0^r|^2}{2\eta_1}$$

Power Conservation for Incident, Reflected, and Transmitted Waves

- Assume an incident wave in medium 1.
- Consider a spot of area *A* on the surface.
- Let's apply power conservation to this spot.

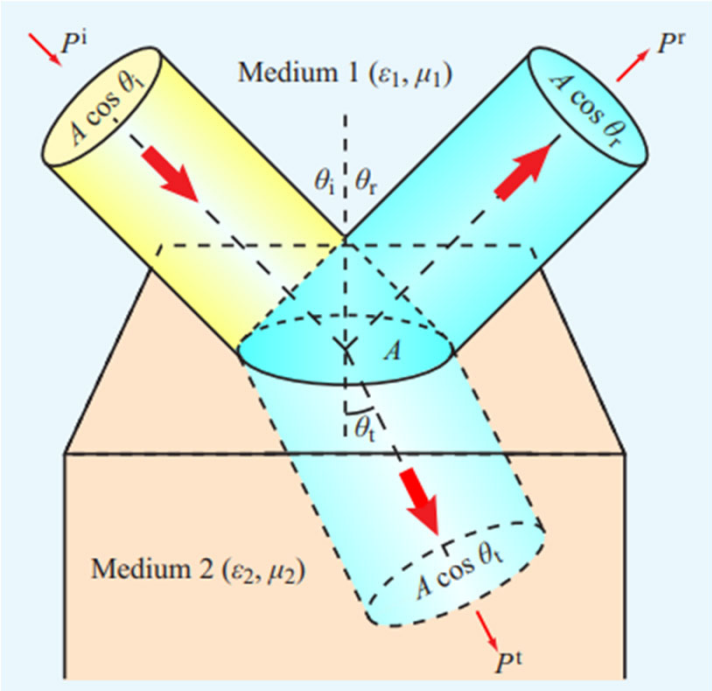

Figure 8-18 Reflection and transmission of an incident circular beam illuminating a spot of size A on the interface.

Power Conservation for Incident, Reflected, and Transmitted Waves (1 of 2)

- Imagine a small cylindrical pill box with a top of area *A* in medium 1, a bottom of area *A* in medium 2, and negligible height, and hence, negligible area of the side.
- Let's consider the net power entering the box.
- The normal to the top surface is in the +z-direction and the normal to the bottom surface is in the –z-direction.
- Power conservation:

$$\hat{z} \cdot \left(\vec{S}^{i} A + \vec{S}^{r} A\right) - \hat{z} \cdot \vec{S}^{t} A = 0$$

Power Conservation for Incident, Reflected, and Transmitted Waves (2 of 2)

- Hence,

$$\hat{z} \cdot \left(\vec{S}^i A + \vec{S}^r A - \vec{S}^t A\right) = 0 \quad \Rightarrow \quad P^i - P^r - P^t = 0$$

$$\boxed{P^i - P^r = P^t}$$

- Note that

$$\hat{z} \cdot \hat{k}_i = \cos\theta_i$$

$$\hat{z} \cdot \hat{k}_r = -\cos\theta_r = -\cos\theta_i$$

$$\hat{z} \cdot \hat{k}_t = \cos\theta_t$$

Reflectivity (Reflectance)

- The reflectivity (reflectance) is the ratio of the power in the reflected wave to the power in the incident wave:

$$R = \frac{P^r}{P^i} = \frac{S^r A \cos\theta_r}{S^i A \cos\theta_i} = \frac{S^r}{S^i} = \frac{\left|E_0^r\right|^2}{\left|E_0^i\right|^2} = \left|\Gamma\right|^2$$

Transmissivity (Transmittance)

- The transmissivity (transmittance) is the ratio of the power in the transmitted wave to the power in the incident wave:

$$T = \frac{P^t}{P^i} = \frac{S^t A \cos\theta_t}{S^i A \cos\theta_i} = \frac{S^t \cos\theta_t}{S^i \cos\theta_i} = \begin{cases} \dfrac{\left|E_0^t\right|^2 \eta_1 \cos\theta_t}{\left|E_0^i\right|^2 \eta_2 \cos\theta_i} = \left|\tau\right|^2 \dfrac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}, & \theta_i < \theta_c \\ 0, & \theta_i \geq \theta_c \end{cases}$$

Power Conservation

- From power conservation:

$$P^i - P^r = P^t$$

$$P^r + P^t = P^i \quad \Rightarrow \quad \frac{P^r}{P^i} + \frac{P^t}{P^i} = 1$$

$$R + T = 1$$

Summary

Table 8-2 Expressions for Γ, τ, R, and T for wave incidence from a medium with intrinsic impedance η_1 onto a medium with intrinsic impedance η_2. Angles θ_i and θ_t are the angles of incidence and transmission, respectively.

Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \dfrac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_\perp = \dfrac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\Gamma_\parallel = \dfrac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
Transmission coefficient	$\tau = \dfrac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_\perp = \dfrac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\tau_\parallel = \dfrac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_\perp = 1 + \Gamma_\perp$	$\tau_\parallel = (1 + \Gamma_\parallel)\dfrac{\cos\theta_i}{\cos\theta_t}$
Reflectivity	$R = \lvert\Gamma\rvert^2$	$R_\perp = \lvert\Gamma_\perp\rvert^2$	$R_\parallel = \lvert\Gamma_\parallel\rvert^2$
Transmissivity	$T = \lvert\tau\rvert^2 \left(\dfrac{\eta_1}{\eta_2}\right)$	$T_\perp = \lvert\tau_\perp\rvert^2 \dfrac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$	$T_\parallel = \lvert\tau_\parallel\rvert^2 \dfrac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$
Relation of R to T	$T = 1 - R$	$T_\perp = 1 - R_\perp$	$T_\parallel = 1 - R_\parallel$

Notes: (1) $\sin\theta_t = \sqrt{\mu_1\varepsilon_1/\mu_2\varepsilon_2}\sin\theta_i$; (2) $\eta_1 = \sqrt{\mu_1/\varepsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2/\varepsilon_2}$; (4) for nonmagnetic media, $\eta_2/\eta_1 = n_1/n_2$.

Equations for transmissivity are valid for $\theta_i < \theta_c$

$T = 0$ for $\theta_i \geq \theta_c$

Question

- Consider a parallel-polarized plane wave traveling in a non-magnetic dielectric medium with dielectric constant of 4 arriving at a planar interface with free space at an angle of 60 degrees wrt the normal. Compute reflection coefficient, transmission coefficient, reflectivity, and transmittivity.

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